

Student Exam Results – PCA Analysis

Equipo 7

Integrantes:

Fajardo Zavala Yendali Viridiana
González Rivas José Rafael
Maldonado Zamora Silvia Viridiana
Olivera Flores Jonathan
Valencia Quiroz Luis Roberto

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Descripción

La base **Student_Exam_Results.csv** contiene calificaciones de exámenes de 1000 estudiantes en seis asignaturas.

Incluye tres columnas adicionales:

- **Total** (suma de todas las calificaciones, debe eliminarse)
 - **Results**: 0 = no aprobado, 1 = aprobado
 - **Div**: categorías 0, 1, 2, 3 (significado no claro)
-

Librerías

```
#library(learnr)  
library(tidyverse)  
library(tidymodels)  
library(embed)  
library(ggcorrplot)  
library(corrr)  
library(tidytext)  
#library(gradethis)  
library(psych)  
#library(sortable)  
#library(learntidymodels)  
library(rstatix)  
library(broom)  
library(rgl)  
library(plotly)  
library(GGally)
```

```
library(tidyr)
library(FactoMineR)
library(factoextra)
library(ggord)

theme_set(theme_bw(16))
```

Importación de Datos

```
df <- read.csv("C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Student_Results/Student_Exam_Resu
dim(df)
```

```
## [1] 1000    9
```

```
glimpse(df)
```

```
## Rows: 1,000
## Columns: 9
## $ Hindi      <int> 65, 94, 7, 88, 39, 51, 73, 18, 15, 53, 61, 49, 21, 93, 44, 5~
## $ English    <int> 10, 56, 85, 46, 81, 43, 48, 23, 64, 85, 17, 10, 24, 57, 6, 5~
## $ Science    <int> 59, 4, 76, 59, 37, 14, 6, 97, 14, 96, 16, 81, 83, 6, 72, 39,~
## $ Maths      <int> 3, 67, 99, 94, 38, 53, 38, 65, 43, 76, 16, 13, 93, 94, 59, 9~
## $ History    <int> 71, 91, 60, 52, 6, 64, 50, 15, 59, 83, 20, 39, 93, 23, 40, 5~
## $ Geograpgy  <int> 37, 50, 25, 38, 54, 59, 21, 20, 59, 62, 48, 86, 28, 84, 6, 4~
## $ Total      <int> 245, 362, 352, 377, 255, 284, 236, 238, 254, 455, 178, 278, ~
## $ Results    <int> 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, ~
## $ Div        <int> 3, 1, 2, 1, 3, 2, 3, 3, 3, 1, 0, 2, 2, 2, 3, 2, 3, 2, 2, 2, ~
```

```
head(df)
```

```
##   Hindi English Science Maths History Geograpgy Total Results Div
## 1    65     10      59     3      71         37   245        0    3
## 2    94     56       4    67     91         50   362        1    1
## 3     7     85      76    99     60         25   352        0    2
## 4    88     46      59    94     52         38   377        1    1
## 5    39     81      37    38      6         54   255        1    3
## 6    51     43      14    53     64         59   284        1    2
```

Preparación de Datos

Selección de columnas relevantes

```
df1 <- df[,1:9]
head(df1)
```

```
##      Hindi English Science Maths History Geograpgy Total Results Div
## 1      65      10      59      3      71      37    245      0    3
## 2      94      56       4      67      91      50    362      1    1
## 3       7      85      76      99      60      25    352      0    2
## 4      88      46      59      94      52      38    377      1    1
## 5      39      81      37      38       6      54    255      1    3
## 6      51      43      14      53      64      59    284      1    2
```

Valores faltantes

```
colSums(is.na(df1))
```

```
##      Hindi   English   Science   Maths   History Geograpgy   Total   Results
##         0         0         0         0         0         0         0         0
##      Div
##         0
```

```
summary(df1)
```

```
##      Hindi      English      Science      Maths
## Min.   : 0.00   Min.   : 0.00   Min.   : 0.00   Min.   : 0.00
## 1st Qu.:26.00   1st Qu.:26.00   1st Qu.:25.00   1st Qu.:25.75
## Median :53.00   Median :50.50   Median :50.00   Median :49.00
## Mean   :51.65   Mean   :50.11   Mean   :49.44   Mean   :49.55
## 3rd Qu.:77.00   3rd Qu.:75.00   3rd Qu.:73.25   3rd Qu.:74.00
## Max.   :99.00   Max.   :99.00   Max.   :99.00   Max.   :99.00
##      History      Geograpgy      Total      Results      Div
## Min.   : 0.00   Min.   : 0.00   Min.   :103.0   Min.   :0.00   Min.   :0.000
## 1st Qu.:24.00   1st Qu.:26.00   1st Qu.:254.0   1st Qu.:0.00   1st Qu.:1.000
## Median :49.00   Median :49.00   Median :296.0   Median :0.00   Median :2.000
## Mean   :49.03   Mean   :50.03   Mean   :299.8   Mean   :0.35   Mean   :1.856
## 3rd Qu.:73.25   3rd Qu.:75.00   3rd Qu.:349.2   3rd Qu.:1.00   3rd Qu.:3.000
## Max.   :99.00   Max.   :99.00   Max.   :505.0   Max.   :1.00   Max.   :3.000
```

Eliminación de la columna Total

```
df1 <- df1 %>% select(-Total)
```

Conversión a factores

```
df1 <- df1 %>%
  mutate(
    Results = as.factor(Results),
    Div = as.factor(Div)
  )

count(df1, Results)
```

```
##   Results   n
## 1      0 650
## 2      1 350
```

```
count(df1, Div)
```

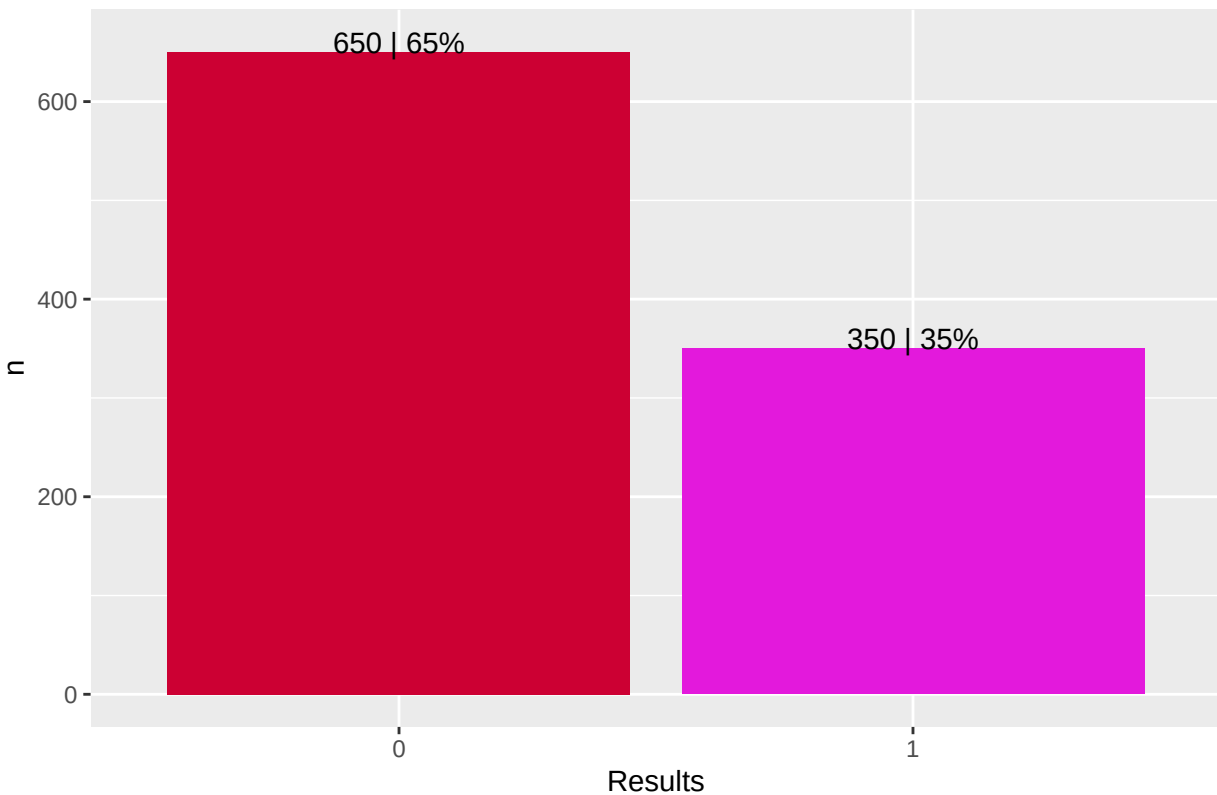
```
##   Div   n
## 1   0  95
## 2   1 206
## 3   2 447
## 4   3 252
```

Distribuciones de variables objetivo

Distribución de Results

```
df1 %>%
  group_by(Results) %>%
  summarise(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = Results, y = n)) +
  geom_col(fill = c("#CC0033", "#e319dc")) +
  geom_text(
    aes(label = paste0(n, " | ", signif(prop * 100, digits = 4), "%")),
    nudge_y = 10
  ) +
  ggtitle("Distribución de Resultados (No Aprobado vs Aprobado)") +
  theme_gray()
```

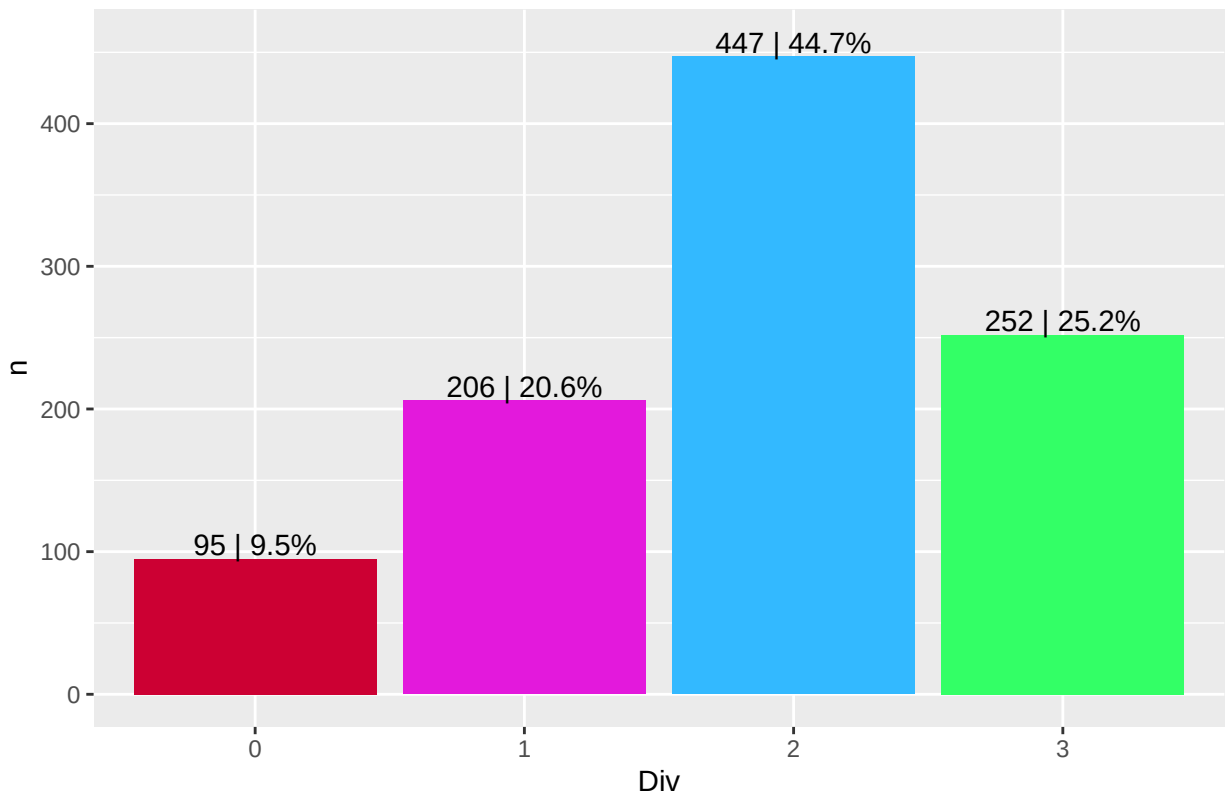
Distribución de Resultados (No Aprobado vs Aprobado)



Distribución de Div

```
df1 %>%
  group_by(Div) %>%
  summarise(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = Div, y = n)) +
  geom_col(fill = c("#CC0033", "#e319dc", "#33b9ff", "#33ff66")) +
  geom_text(
    aes(label = paste0(n, " | ", signif(prop * 100, digits = 4), "%")),
    nudge_y = 10
  ) +
  ggtitle("Distribución de Divisiones (0, 1, 2, 3)") +
  theme_gray()
```

Distribución de Divisiones (0, 1, 2, 3)



Análisis Exploratorio

Medias de variables numéricas

```
df1 %>%
  select(where(is.numeric)) %>%
  colMeans()
```

```
##      Hindi   English   Science   Maths   History Geograpgy
##    51.645    50.110    49.440    49.553    49.034    50.028
```

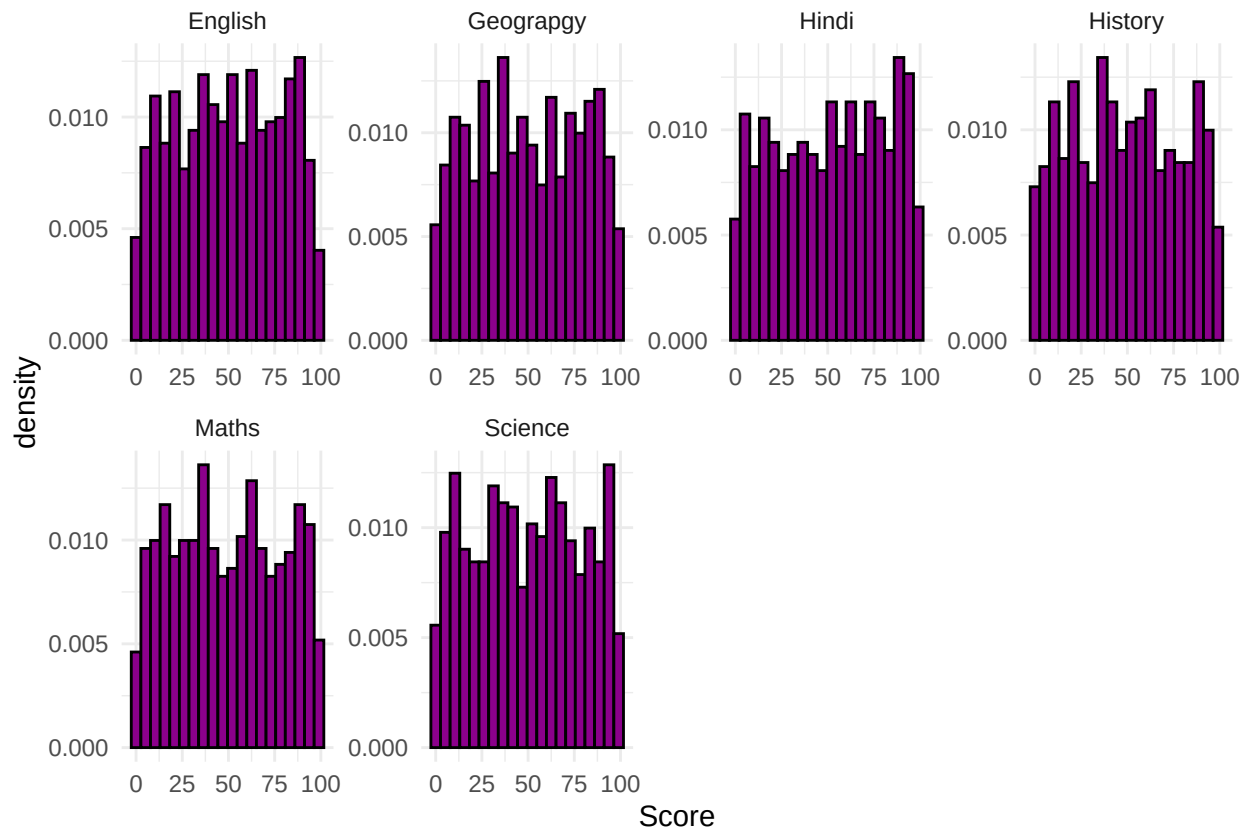
Histogramas por variable

```
df1 %>%
  pivot_longer(
```

```

cols = !c(Results, Div),
names_to = "Variable",
values_to = "Score"
) %>%
ggplot(aes(x = Score)) +
geom_histogram(aes(y = ..density..), bins = 20,
               colour = "black", fill = "darkmagenta") +
facet_wrap(~ Variable, ncol = 4, scales = "free") +
theme_minimal()

```



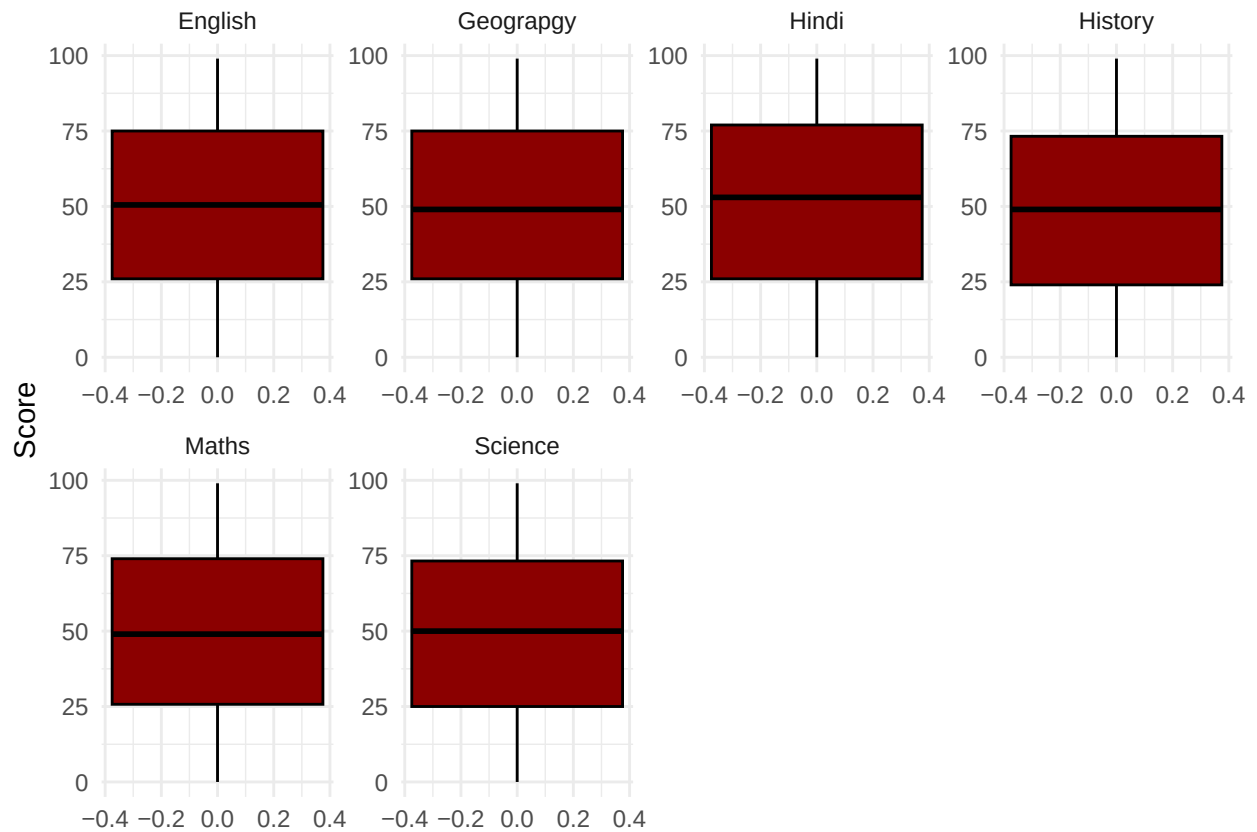
Boxplots por variable

```

df1 %>%
pivot_longer(
  cols = !c(Results, Div),
  names_to = "Variable",
  values_to = "Score"
) %>%
ggplot(aes(y = Score)) +
geom_boxplot(fill = "darkred", color = "black") +

```

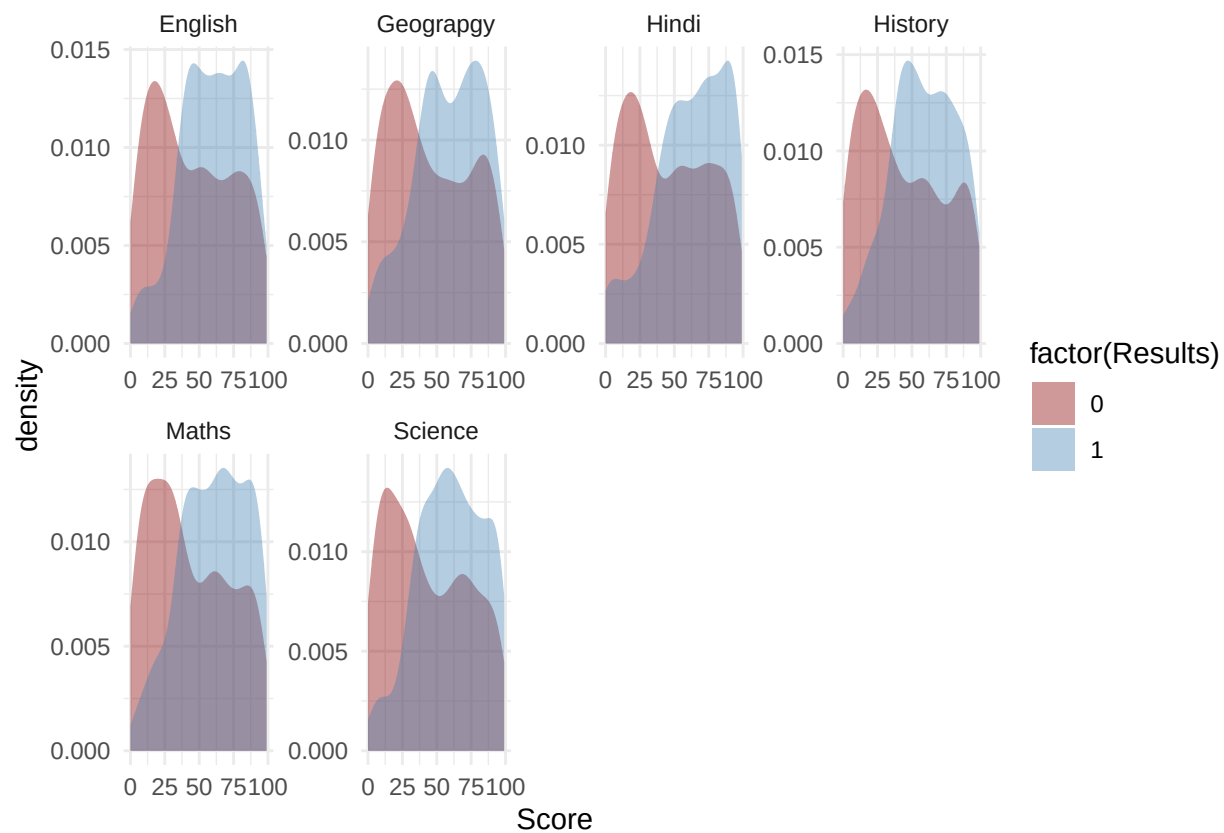
```
facet_wrap(~ Variable, ncol = 4, scales = "free") +
theme_minimal()
```



Gráficas de densidad

Por Results

```
df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score, fill = factor(Results))) +
  geom_density(alpha = 0.4, colour = NA) +
  facet_wrap(~ Variable, ncol = 4, scales = "free") +
  scale_fill_manual(values = c("0"="darkred", "1"="steelblue")) +
  theme_minimal()
```

Por Div

```
df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score, fill = factor(Div))) +
  geom_density(alpha = 0.4, colour = NA) +
  facet_wrap(~ Variable, ncol = 4, scales = "free") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```



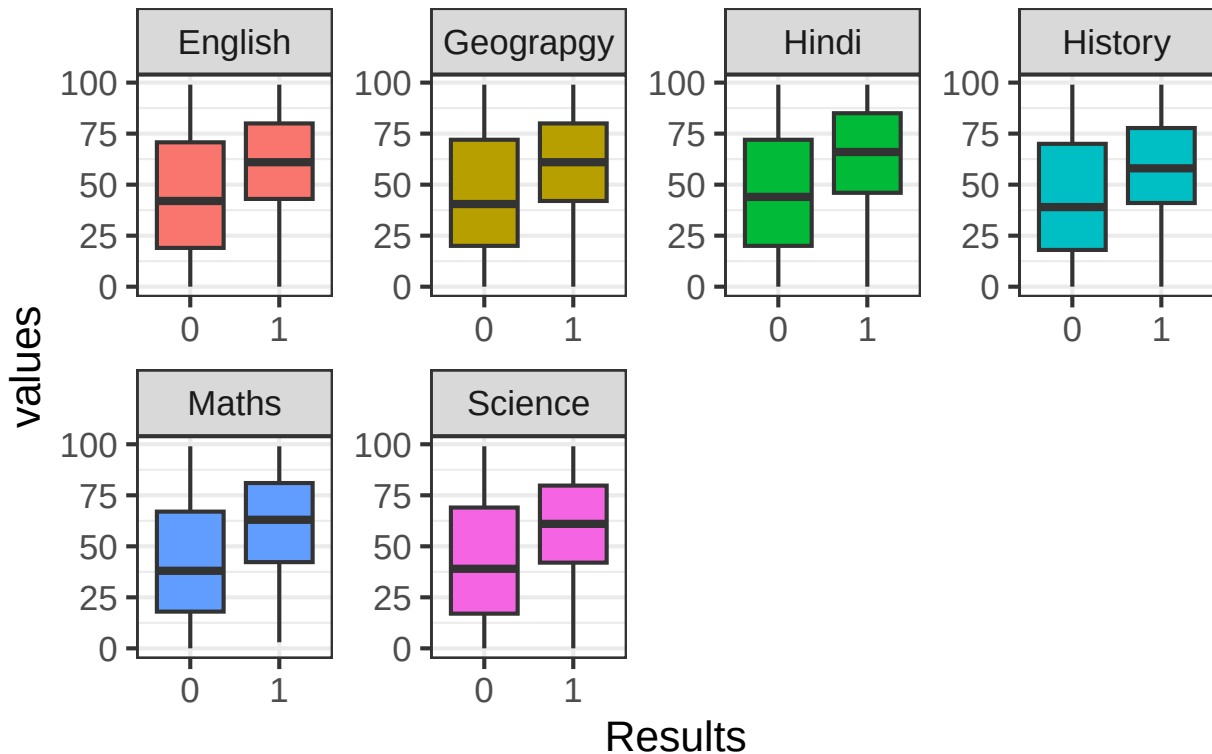
Comparación por variable de respuesta (Results)

```
df1_long <- df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "predictores", values_to = "values")
```

Boxplots

```
df1_long %>%
  ggplot(aes(x = Results, y = values, fill = predictores)) +
  geom_boxplot() +
  facet_wrap(~ predictores, scales = "free", ncol = 4) +
  theme(legend.position = "none") +
  labs(title = "Comparación vía box-plot")
```

Comparación vía box-plot

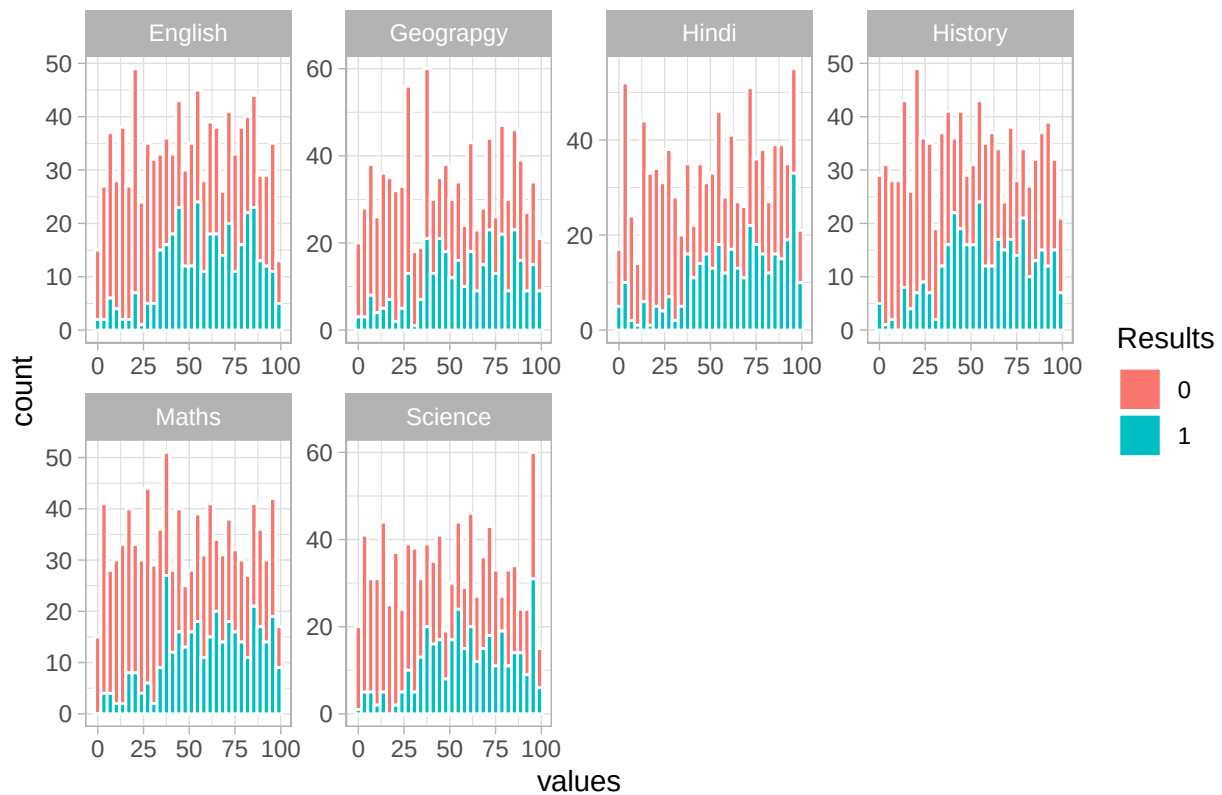


Histogramas

```
df1_long %>%  
  ggplot(aes(values, fill = Results)) +  
  geom_histogram(color = "white") +  
  facet_wrap(~ predictores, scales = "free", ncol=4) +  
  theme_light() +  
  labs(title = "Comparación vía histograma")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

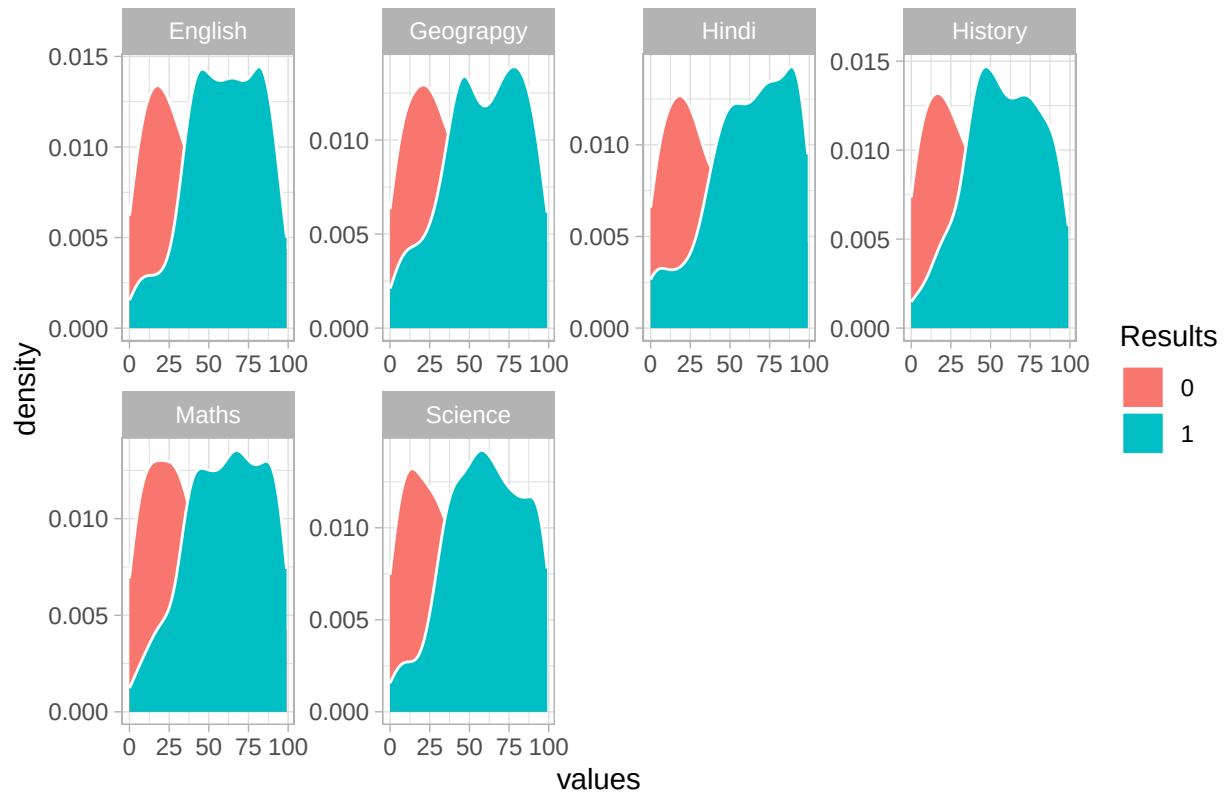
Comparación vía histograma



Densidad

```
df1_long %>%  
  ggplot(aes(values, fill = Results)) +  
  geom_density(color = "white") +  
  facet_wrap(~ predictores, scales = "free", ncol=4) +  
  theme_light() +  
  labs(title = "Comparación a través de densidad")
```

Comparación a través de densidad



Matriz de correlación

```
grades_corr <- df1 %>%
  select(-Results, -Div) %>%
  correlate(method = "spearman")
```

```
## Correlation computed with
## * Method: 'spearman'
## * Missing treated using: 'pairwise.complete.obs'
```

```
grades_corr
```

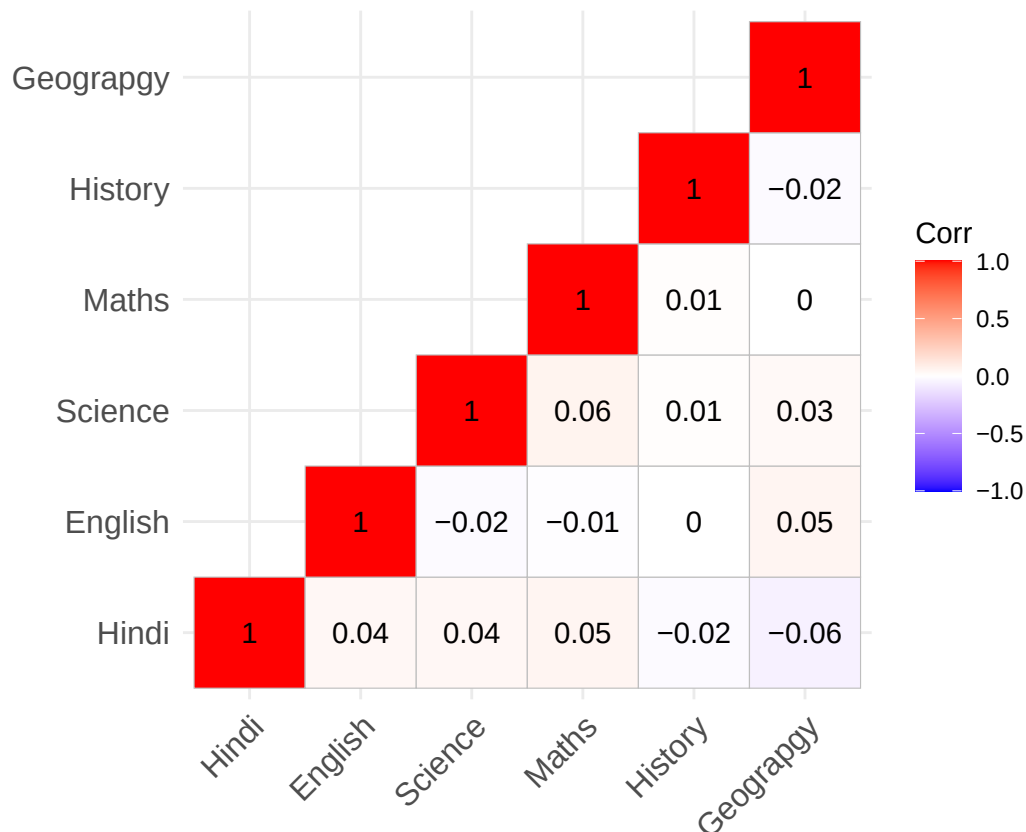
```
## # A tibble: 6 x 7
##   term      Hindi English Science Maths History Geograpy
##   <chr>    <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
## 1 Hindi    NA      0.0370  0.0387  0.0454 -0.0212 -0.0633
## 2 English  0.0370 NA      -0.0234 -0.0131 -0.00114 0.0523
## 3 Science  0.0387 -0.0234 NA      0.0559  0.00565 0.0285
## 4 Maths    0.0454 -0.0131 0.0559 NA      0.0121 -0.00115
## 5 History -0.0212 -0.00114 0.00565 0.0121 NA      -0.0209
## 6 Geograpy -0.0633 0.0523 0.0285 -0.00115 -0.0209 NA
```

Correlation matrix (base R)

```
corr_stg <- cor(df1[, -c(7, 8)], method = "spearman")
corr_stg
```

```
##           Hindi      English      Science      Maths      History
## Hindi      1.00000000  0.036962705  0.038719755  0.045364412 -0.021182424
## English    0.03696271  1.000000000 -0.023392589 -0.013091984 -0.001138727
## Science    0.03871975 -0.023392589  1.000000000  0.055931481  0.005645743
## Maths      0.04536441 -0.013091984  0.055931481  1.000000000  0.012092849
## History    -0.02118242 -0.001138727  0.005645743  0.012092849  1.000000000
## Geograpgy -0.06325747  0.052349020  0.028464649 -0.001147157 -0.020918686
##           Geograpgy
## Hindi      -0.063257467
## English     0.052349020
## Science     0.028464649
## Maths      -0.001147157
## History    -0.020918686
## Geograpgy  1.000000000
```

```
ggcorrplot::ggcorrplot(corr = corr_stg,
                        type = "lower",
                        show.diag = TRUE,
                        lab = TRUE)
```



Determinante

```
det(corr_stg)
```

```
## [1] 0.9822475
```

KMO

```
psych::KMO(corr_stg)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: psych::KMO(r = corr_stg)
## Overall MSA = 0.48
## MSA for each item =
##      Hindi   English   Science   Maths   History Geograpgy
##      0.48     0.46     0.49     0.52     0.46     0.47
```

Test de Bartlett

```
psych::cortest.bartlett(corr_stg, n = dim(df1)[1])
```

```
## $chisq
## [1] 17.84333
##
## $p.value
## [1] 0.2709919
##
## $df
## [1] 15
```

Preparación para PCA

```
pca_recipe <- recipe(~., data = df1) %>%
  update_role(Results, new_role = "id") %>%
  update_role(Div, new_role = "id") %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_predictors()) %>%
  step_pca(all_numeric_predictors(), threshold = 0.80, id="pca")
```

```
pca_recipe
```

```
##
```

```
## -- Recipe -----
##
## -- Inputs

## Number of variables by role

## predictor: 6
## id:        2

##

## -- Operations

## * Dummy variables from: all_nominal_predictors()

## * Centering and scaling for: all_predictors()

## * PCA extraction with: all_numeric_predictors()
```

Ejecución del PCA

```
stg_pca <- prep(pca_recipe)
```

Loadings

```
pca_loading <- tidy(stg_pca, id="pca")
pca_loading
```

```
## # A tibble: 36 x 4
##   terms      value component id
##   <chr>      <dbl> <chr>   <chr>
## 1 Hindi    -0.583   PC1     pca
## 2 English   0.133   PC1     pca
## 3 Science  -0.482   PC1     pca
## 4 Maths    -0.551   PC1     pca
## 5 History  -0.0228  PC1     pca
## 6 Geograpgy 0.326    PC1     pca
## 7 Hindi     0.000405 PC2     pca
## 8 English  -0.483   PC2     pca
## 9 Science  -0.380   PC2     pca
## 10 Maths   -0.223   PC2     pca
## # i 26 more rows
```


Varianzas

```
pca_variances <- tidy(stg_pca, id="pca", type="variance")
pca_variances
```

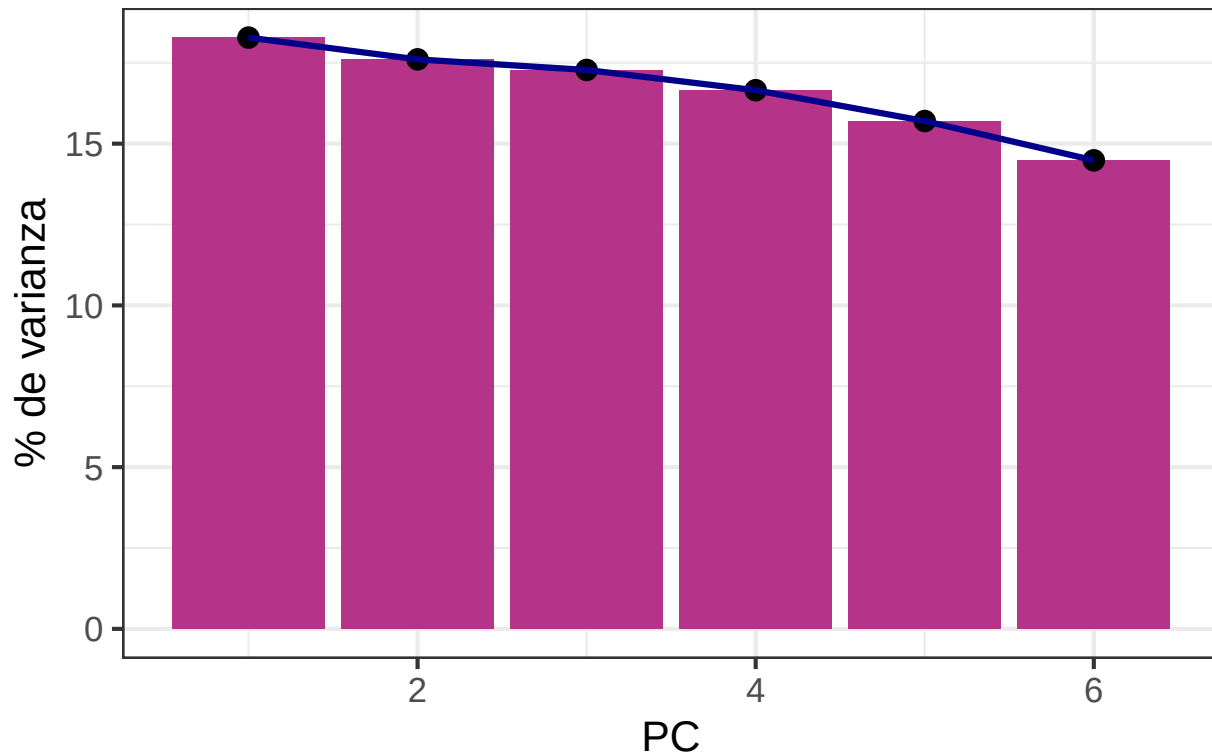
```
## # A tibble: 24 x 4
##   terms          value component id
##   <chr>         <dbl>     <int> <chr>
## 1 variance      1.10         1 pca
## 2 variance      1.06         2 pca
## 3 variance      1.04         3 pca
## 4 variance      0.999        4 pca
## 5 variance      0.942        5 pca
## 6 variance      0.869        6 pca
## 7 cumulative variance 1.10         1 pca
## 8 cumulative variance 2.15         2 pca
## 9 cumulative variance 3.19         3 pca
## 10 cumulative variance 4.19         4 pca
## # i 14 more rows
```

Scree plots

% de varianza por componente

```
stg_pca %>%
  tidy(id="pca", type="variance") %>%
  filter(terms == "percent variance") %>%
  ggplot(aes(x = component, y = value)) +
  geom_col(fill="#B53389") +
  geom_point(size=3) +
  geom_line(color="darkblue", size=1.1) +
  labs(x="PC", y="% de varianza", title="Scree Plot")
```

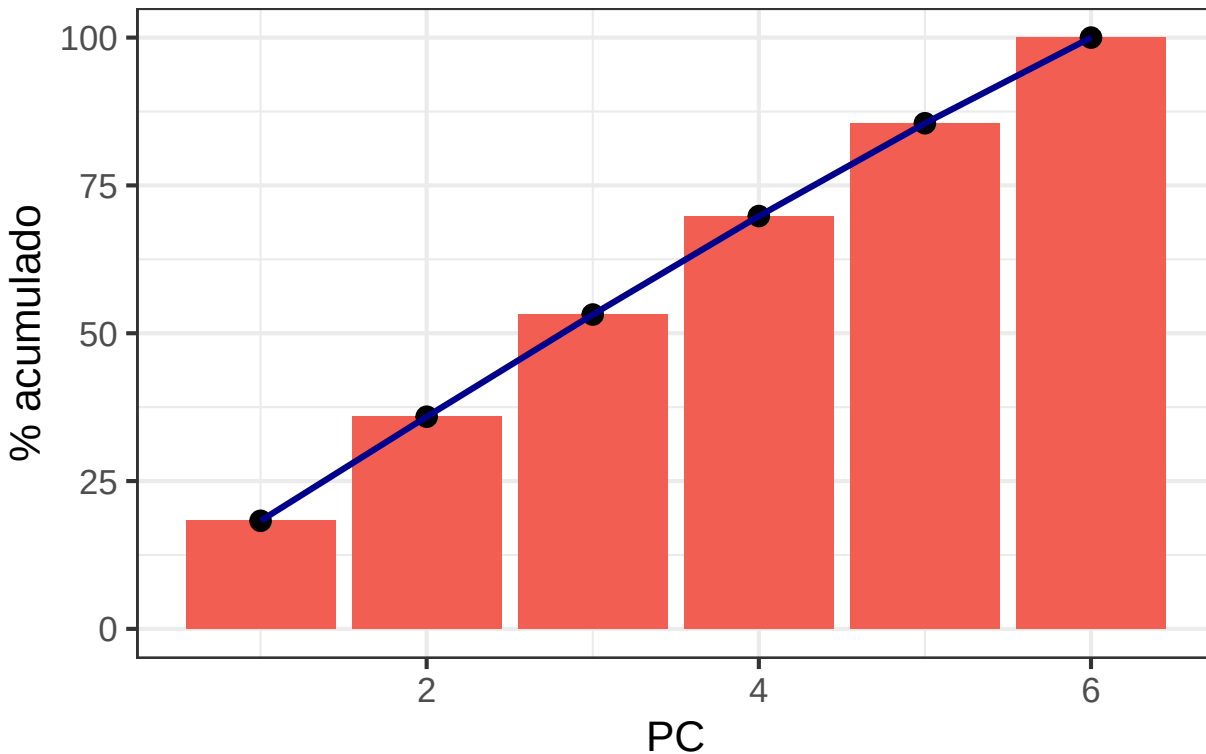
Scree Plot



% acumulado de varianza

```
stg_pca %>%  
  tidy(id="pca", type="variance") %>%  
  filter(terms == "cumulative percent variance") %>%  
  ggplot(aes(x = component, y = value)) +  
  geom_col(fill="#F25E52") +  
  geom_point(size=3) +  
  geom_line(color="darkblue", size=1.1) +  
  labs(x="PC", y="% acumulado", title="Scree Plot Acumulado")
```

Scree Plot Acumulado



Varianza con 2 variables solamente

```
variance_exp <- tidy(stg_pca, id = "pca", type = "variance") %>%
  filter(str_detect(terms, "percent variance"))
variance_exp
```

```
## # A tibble: 12 x 4
##   terms                value component id
##   <chr>                <dbl>     <int> <chr>
## 1 percent variance      18.3         1 pca
## 2 percent variance      17.6         2 pca
## 3 percent variance      17.3         3 pca
## 4 percent variance      16.7         4 pca
## 5 percent variance      15.7         5 pca
## 6 percent variance      14.5         6 pca
## 7 cumulative percent variance 18.3         1 pca
## 8 cumulative percent variance 35.9         2 pca
## 9 cumulative percent variance 53.2         3 pca
##10 cumulative percent variance 69.8         4 pca
##11 cumulative percent variance 85.5         5 pca
##12 cumulative percent variance 100          6 pca
```

Plot con solo 2 variables

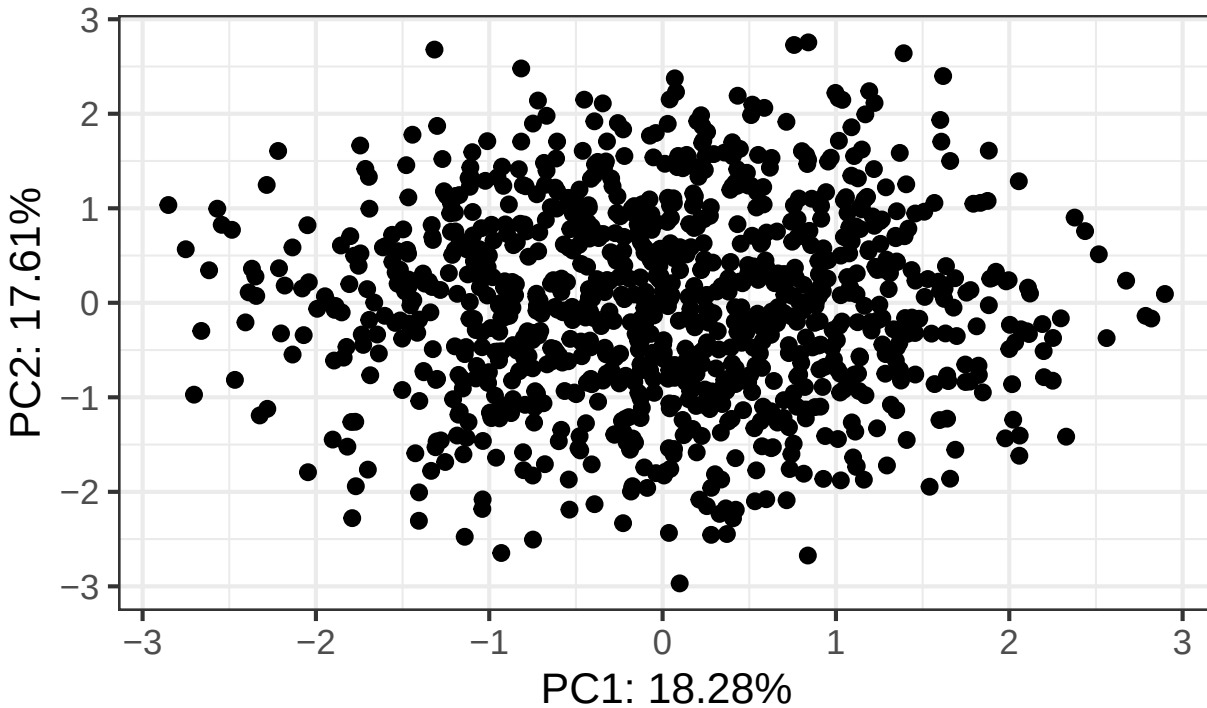
```
datos_pca<-bake(stg_pca, new_data = NULL)
datos_pca
```

```
## # A tibble: 1,000 x 7
##   Results Div    PC1    PC2    PC3    PC4    PC5
##   <fct>   <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0       3     0.117 1.42   0.477 -0.223 1.67
## 2 1       1    -0.422 0.680 -0.767 1.67  -0.886
## 3 0       2    -0.639 -0.620 0.854  0.969 -0.977
## 4 1       1    -1.89  -0.0746 -0.157 0.222 -0.706
## 5 1       3     0.905 -0.706 -1.18  -0.743 -0.297
## 6 1       2     0.593 0.448  -0.0000810 0.456 -0.752
## 7 0       3     0.183 1.44   -1.22   0.212 -0.423
## 8 0       3    -0.867 0.220  1.20   -1.64  0.198
## 9 0       3     1.60  0.126  0.106  0.578 -0.762
## 10 1      1    -1.04 -1.46  0.659  1.51  0.450
## # i 990 more rows
```

```
datos_pca %>%
  ggplot(aes(x = PC1, y = PC2))+
  geom_point()+
  labs(x = paste0("PC1: ",round(variance_exp$value[[1]],2), "%"),
       y = paste0("PC2: ",round(variance_exp$value[[2]],2), "%"))+
  ggtitle(paste0("STD TGR: Gráfica de componentes principales","\n",VE))
```

STDTGR: Gráfica de componentes principales

Varianza explicada por dos componentes: 35.89 %



Conclusiones

- Solo se retiene alrededor del **85%** de la información con 5 componentes.
 - La correlación entre variables es **muy baja**, por lo que PCA no es la mejor técnica, pero se realizó por motivos de práctica.
-